

SOC L1 INCIDENT REPORT

Authentication Attack & Suspected Privilege Escalation

Report Type	Security Incident / Authentication Investigation
Prepared By	SOC Level 1 Analyst
Analysis Platform	Splunk Enterprise
Data Source	auth.log
Splunk Index	soc_logs
Host	LAPTOP-7ACAMDT4
Detection Date	23 August 2026
Analyst Assessment	High Priority – Requires L2 / Incident Response Investigation

Confidentiality: Internal SOC use only. This report contains security-event details and should be handled according to the organisation's incident-response and information-classification procedures.

1. Executive Summary

A review of authentication events in **auth.log** identified a suspicious sequence involving the **admin** account from source IP **185.220.101.45**. The activity consisted of five consecutive failed login attempts, followed by a successful login and, approximately three minutes later, a recorded **PRIVILEGE_ESCALATION** event. The sequence is consistent with a potential credential-attack scenario followed by successful account access and possible privilege escalation.

The activity should **not be treated as confirmed compromise solely from these log entries**. However, the combination of repeated failures, subsequent successful authentication, and privilege escalation is sufficiently suspicious to warrant escalation to an L2 analyst / incident response team for validation, scope determination, and containment decisions.

Primary incident hypothesis: An external source attempted repeated authentication against the admin account, eventually succeeded, and subsequently generated a privilege-escalation event.

Incident Classification

Field	Assessment
Incident Type	Suspicious authentication activity / possible account compromise
Severity	HIGH – successful authentication followed by privilege escalation
Confidence	Medium – strong event sequence, but underlying authentication and escalation cause require validation
Affected Account	admin
Primary Source IP	185.220.101.45
Observed Actions	LOGIN_FAILED ×5 → LOGIN_SUCCESS → PRIVILEGE_ESCALATION
Data Source	auth.log / Splunk index soc_logs

2. Scope and Evidence

The investigation was performed using Splunk Enterprise against the uploaded **auth.log** file indexed under **soc_logs**. The dashboard created during the investigation provides visibility into total security events and events grouped by action.

Evidence Observed

Evidence	Value / Observation
Log file	auth.log
Splunk index	soc_logs
Host	LAPTOP-7ACAMDT4
Total events ingested	13
LOGIN_FAILED events	8 total
LOGIN_SUCCESS events	3 total
PRIVILEGE_ESCALATION events	1 total
Primary suspicious source	185.220.101.45

3. Incident Timeline

All timestamps below are based on the timestamps visible in the supplied authentication log.

Time	Account	Source IP	Action	SOC Interpretation
08:15:21	admin	192.168.1.10	LOGIN_SUCCESS	Internal successful admin authentication
08:22:14	alen	192.168.1.15	LOGIN_SUCCESS	Internal successful user authentication
09:01:33	admin	185.220.101.45	LOGIN_FAILED	External failed authentication
09:01:35	admin	185.220.101.45	LOGIN_FAILED	Repeated failed authentication
09:01:38	admin	185.220.101.45	LOGIN_FAILED	Repeated failed authentication
09:01:41	admin	185.220.101.45	LOGIN_FAILED	Repeated failed authentication
09:01:44	admin	185.220.101.45	LOGIN_FAILED	Fifth consecutive failure
09:02:10	admin	185.220.101.45	LOGIN_SUCCESS	Successful authentication after failures
09:05:22	admin	185.220.101.45	PRIVILEGE_ESCALATION	Privilege escalation recorded; requires validation
10:11:45	alen	192.168.1.15	LOGIN_SUCCESS	Later internal successful authentication
11:23:12	guest	203.0.113.50	LOGIN_FAILED	Failed guest authentication
11:23:15	guest	203.0.113.50	LOGIN_FAILED	Repeated guest failure
11:23:18	guest	203.0.113.50	LOGIN_FAILED	Third guest failure

4. SOC L1 Analysis

4.1 Authentication pattern: The most significant sequence is the five failed login attempts against the admin account from 185.220.101.45 between 09:01:33 and 09:01:44, followed by a successful login at 09:02:10. The short interval between failures and success increases the likelihood that the events are related.

4.2 Account risk: The targeted account is **admin**, which is inherently high-value because administrative accounts generally have elevated privileges. A successful authentication to such an account should therefore receive greater scrutiny than an ordinary user login.

4.3 Privilege escalation: At 09:05:22, the same source IP generated a PRIVILEGE_ESCALATION event for the admin account. This is the strongest indicator in the available evidence. The log does not reveal the mechanism used for escalation, so L1 should avoid asserting the exact exploit or technique until additional endpoint/system telemetry is reviewed.

4.4 Other activity: A separate source, 203.0.113.50, generated three failed guest logins. This is suspicious but less critical than the admin sequence. The address 203.0.113.0/24 is reserved for documentation/testing, so this event may represent lab-generated data rather than a routable external attacker. It should still be retained as part of the detection scenario.

5. Indicators of Interest

Type	Indicator	Relevance
Source IP	185.220.101.45	Primary suspicious source; failed logins → success → privilege escalation
Account	admin	High-value account targeted in suspicious sequence
Source IP	203.0.113.50	Three failed guest logins; likely lab/documentation traffic
Internal IP	192.168.1.10	Successful admin login; internal/private address
Internal IP	192.168.1.15	Successful alen logins; internal/private address
Host	LAPTOP-7ACAMDT4	Host associated with ingested events

6. Splunk Detection Queries Used

The following searches were used during the investigation and can be retained as repeatable SOC detection logic.

Failed authentication volume by source:

```
index=soc_logs action="LOGIN_FAILED" | stats count by src_ip
```

Focused investigation of the suspicious source:

```
index=soc_logs src_ip="185.220.101.45" | table _time username src_ip action | sort _time
```

Action distribution:

```
index=soc_logs | stats count by action
```

7. MITRE ATT&CK; Alignment

T1110 – Brute Force: The repeated failed authentication attempts are consistent with a brute-force or password-guessing pattern. The exact sub-technique cannot be established from the available log alone.

T1078 – Valid Accounts: The successful admin authentication after repeated failures may indicate use of valid credentials. This is an assessment, not proof of credential theft.

Privilege Escalation: A privilege-escalation action is explicitly recorded in the source log. The specific ATT&CK; technique should be assigned only after endpoint/process/audit evidence identifies how the escalation occurred.

8. Potential Impact Assessment

If the admin authentication was unauthorized, the potential impact is significant because an attacker could have obtained administrative access. Possible consequences include modification of security controls, access to sensitive resources, creation or modification of accounts, persistence, lateral movement, or data access. None of these impacts are confirmed by the current evidence.

Current confirmed facts: repeated failed admin authentication, subsequent successful admin authentication from the same source, and a subsequent privilege-escalation event. **Not confirmed:** data theft, malware execution, persistence, lateral movement, or actual system compromise.

9. Recommended L1 → L2 Actions

Priority	Recommended Action	Purpose
P1	Validate whether the admin login at 09:02:10 was authorized.	Determine whether the successful authentication is legitimate.
P1	Investigate the 09:05:22 PRIVILEGE_ESCALATION event using endpoint/security logs to identify the process, user context, and mechanism responsible.	Identify the process, user context, and mechanism responsible.
P1	Review all activity from 185.220.101.45 across available Splunk data sources.	Establish scope and identify additional affected accounts/hosts.
P1	If unauthorized, disable/reset the affected admin credentials and revoke active sessions across all endpoints.	Contain a possible data breach and prevent further compromise.
P2	Check for account creation, group membership changes, scheduled tasks, services, registry changes, and security control changes.	Identify processes and potential system compromise activity.
P2	Review endpoint telemetry around 09:01–09:10.	Correlate authentication with processes and system changes.
P2	Check firewall/VPN/EDR/network logs for the source IP.	Validate origin and determine whether the source interacted with other systems.
P3	Create/retain a Splunk alert for repeated failed logins followed by success.	Improve early detection of password attacks.

10. L1 Conclusion

The event sequence should be escalated as a **high-priority suspicious authentication incident**. The strongest concern is the correlation of five failed admin logins from 185.220.101.45 with a successful admin login shortly afterward and a privilege-escalation event three minutes later. The available evidence supports suspicion of unauthorized access but does not, by itself, prove compromise or identify the escalation mechanism.

L2/IR should validate the legitimacy of the admin login, investigate the privilege-escalation event, correlate additional telemetry, and determine whether containment is required.

Appendix A – Raw Event Sequence

The following is a normalized representation of the events visible in the supplied auth.log screenshot. It is included for auditability and should be compared with the original log if exact forensic evidence is required.

Timestamp	Raw event summary
2026-08-23 08:15:21	username=admin src_ip=192.168.1.10 action=LOGIN_SUCCESS
2026-08-23 08:22:14	username=alen src_ip=192.168.1.15 action=LOGIN_SUCCESS
2026-08-23 09:01:33	username=admin src_ip=185.220.101.45 action=LOGIN_FAILED
2026-08-23 09:01:35	username=admin src_ip=185.220.101.45 action=LOGIN_FAILED
2026-08-23 09:01:38	username=admin src_ip=185.220.101.45 action=LOGIN_FAILED
2026-08-23 09:01:41	username=admin src_ip=185.220.101.45 action=LOGIN_FAILED
2026-08-23 09:01:44	username=admin src_ip=185.220.101.45 action=LOGIN_FAILED
2026-08-23 09:02:10	username=admin src_ip=185.220.101.45 action=LOGIN_SUCCESS
2026-08-23 09:05:22	username=admin src_ip=185.220.101.45 action=PRIVILEGE_ESCALATION
2026-08-23 10:11:45	username=alen src_ip=192.168.1.15 action=LOGIN_SUCCESS
2026-08-23 11:23:12	username=guest src_ip=203.0.113.50 action=LOGIN_FAILED
2026-08-23 11:23:15	username=guest src_ip=203.0.113.50 action=LOGIN_FAILED
2026-08-23 11:23:18	username=guest src_ip=203.0.113.50 action=LOGIN_FAILED

Appendix B – Analyst Notes

The report intentionally distinguishes observed facts from hypotheses. The presence of a successful login after multiple failures is suspicious, but credential compromise cannot be proven from authentication logs alone. Similarly, PRIVILEGE_ESCALATION is an observed event label; its technical cause must be established from endpoint or audit telemetry before attributing it to a specific exploit.

Recommended incident disposition: Escalate to L2/IR for validation and correlation. Close only after the admin activity and privilege escalation have been confirmed as authorized or the incident has been fully investigated.